Design and Implementation of a 6 V Lead-Acid Battery Charger Using a Discrete Series-Pass Regulator with Voltage Feedback

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***Abstract*—This project presents the design, simulation, and implementation of a 6-V lead-acid battery charger using a discrete-series-pass regulator with voltage feedback control. The circuit involves a Zener diode to establish a stable reference voltage, an error amplifier transistor to compare the output with the reference, a power pass transistor operating in the linear region to regulate the charging voltage and current, and a current limiting transistor. Through analytical design, LTSpice simulation, and hardware implementation, the regulator's performance is evaluated under varying input voltages (10–30 V) and load conditions. Key metrics including voltage regulation percentage, output resistance, and efficiency are measured and analyzed. The design emphasizes safe charging practices for lead-acid batteries and incorporates optional enhancements such as printed circuit board (PCB) fabrication. The project provides practical insight into linear regulator design, feedback control principles, and thermal management, while highlighting limitations and potential improvements for future iterations. Overall, the results of testing showed a stabilized voltage output and a proper battery charging under varying input conditions, indicating that the design was integrated and constructed properly.**

***Index Terms*—Voltage Regulator, Load Testing, Efficiency, Voltage Regulation, Lead-Acid Battery, Output Stability, Current Limiting, Input–Output Characteristics, Resistive Loads, Power Dissipation.**

I. Introduction

Battery charging systems play a crucial role in both electronic devices and automotive applications as they link power sources with batteries. These systems must deliver consistent voltage and current to guarantee that batteries function optimally and have an extended lifespan. A variety of techniques, including linear and switching regulators, are employed to maintain a stable output despite variations in input and load conditions. Discrete linear regulators represent a straightforward yet effective solution while illustrating fundamental concepts such as transistor functionality, feedback control, and essential circuit design. Analyzing these systems improves comprehension of power regulation mechanisms, identifies methods to boost efficiency, and guides the creation of reliable charging circuits in real-world applications.

A. Motivation

Lead-acid battery chargers are widely used since they are cheaper than other rechargeable battery types, their mature technology ensures reliability and established manufacturing, Can be designed for high-power (starting) or sustained power (deep cycle), and highly recyclable, supporting a circular economy. However, improper charging practices can greatly diminish the lifespan of a battery, decrease its efficiency, and pose safety hazards, including overheating and sulphation. Moreover, many commercially available batteries lack proper voltage regulation and do not provide sufficiently smooth charging profiles required for long-term battery health. A reliable charging system, therefore, requires stable voltage control, predictable current behavior, and protection against fluctuations in input supply. The project is motivated by the need for a simple, cost-effective and well-regulated 6V lead-acid battery charger that can ensure safe charging conditions while demonstrating core principles of analog power-supply design.

B. Literature Review

The design and analysis of regulated power supplies rely on established principles covering voltage regulation, feedback behavior, and transistor operation. Fundamental concepts of series voltage regulators and regulated power-supply structures from the basis for understanding modern linear regulation techniques [1]. These principles are supported by comprehensive treatments of semiconductor devices and analog circuit behavior, where the interactions among transistors, biasing networks, and small-signal models define regulator performance and stability [2], [3].

More application-oriented works expand on regulator stability, loop compensation, and transient-response optimization, providing guidance for practical implementation of linear regulators in real systems [4]. Additional engineering considerations include battery-charging interfaces, where charging algorithms and cell-dependent behaviors influence power-supply requirements [5], as well as the behavior of pass transistors used in higher-current regulation stages, highlighting

the importance of dropout voltage, thermal limits, and safe-operating-area constraints [6]. Broader discussions on energy-storage technologies further detail the electrical and thermal properties affecting supply design [7].

Control-system methodologies provide the analytical foundation for ensuring regulator stability and robustness, using feedback analysis, state-space modelling, and compensation strategies applicable to power-electronic systems [8]. Simulation tools offer additional support for predicting regulator performance and validating behaviour before hardware implementation [9]. At the component level, long-term behaviour of reference elements such as Zener diodes contributes to precision regulation and noise considerations [10]. Finally, studies of integrated passive technologies emphasize the importance of thermal management, packaging, and parasitic effects in maintaining reliability and efficiency in regulated supplies [11]. Overall, discrete-series-pass regulators with voltage feedback offer precise control of voltage and current for applications like battery charging, and careful analysis of their performance under varying input and load conditions is crucial to ensure efficiency, stability, and safe operation.

II. Materials and Methods

A. Design

A simple voltage regulator is shown in Figure 1; it is constructed by adding a series pass transistor (usually a power transistor), a Zener diode, and a resistor R_S. The load current passes through the series transistor Q_1; that is the reason we call the circuit a series voltage regulator. The use of the Zener diode is to provide a reference voltage for the transistor since a BJT transistor is voltage-controlled. Therefore, the current delivered to R_L will be approximately constant, leading to a regulated voltage across it.

We can see that the output voltage is equal to:

$$V_{out} = V_Z - V_{BE}$$

There are two possible cases that might happen in this circuit:

1) **The output voltage decreases**: Under this circumstance, the base-emitter voltage will increase (since V_Z is constant), causing the transistor to conduct more, which in turn raises the output voltage and restores regulation.
2) **The output voltage increases**: when this occurs, the base-emitter voltage will decrease (since V_Z is constant), causing the transistor to conduct less, decreasing the output voltage back toward its regulated value.

The main advantage of this circuit is that the changes in Zener current are reduced by a factor β. Hence, the effect of the Zener impedance is tremendously reduced, and we will get a more stabilized output. However, a major disadvantage of this circuit is that the output is not absolutely constant even though the Zener current is reduced to a significant amount. This is because the Zener voltage V_Z and the base-emitter voltage V_{BE} are reduced with the rise of temperature, affecting the output voltage and regulation. Another reason is that as

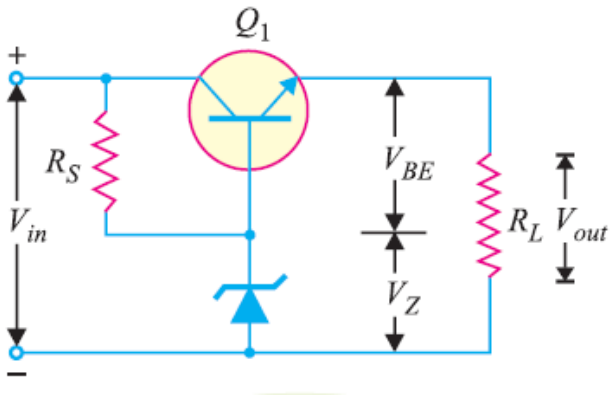


Fig. 1. A simple series regulator.

additional current flows into the base of the transistor, it will decrease as more current is taken by the base of the transistor. Since the Zener voltage experiences only minimal fluctuations despite a significant current change, this minor variation has a small impact on V_{BE} and results in a minor change in the output voltage. Together, these factors can form a significant change in the output voltage when the supply is operating under demanding conditions.

Another feature that can be added to enhance the performance of the regulator is adding an error amplifier transistor as shown in Figure 2. We can see that the added components are an error amplification transistor Q_1, and the resistors R_1, R_2, and R_3. In this circuit, the transistor Q_1 is the error amplifier circuit and its gain is set by the value of its load resistor R_3. The resistors R_1 and R_2 are used as a voltage divider circuit from the output voltage; the relation between them can be shown as an inverse voltage divider:

$$V_{out} = V_F \times \frac{R_1 + R_2}{R_2}$$

Moreover, the reference voltage V_Z is compared with a feedback voltage V_F, which is a portion of the actual output voltage and determined by the following equation:

$$V_F = V_Z + V_{BE(Q_1)}$$

The function of the voltage divider is to sample the output voltage and then deliver a negative feedback voltage to the base of Q_1. The collector current of Q_1 is controlled by the feedback voltage V_F.

There are also two possible cases that might happen in this circuit:

1) **The output voltage increases**: when this happens, the voltage across R_2 will increase. This means that Q_1's base voltage is increased, hence increasing its collector current and the current across R_3. More current across R_3 means that the voltage drop across it will be higher, decreasing the base voltage of Q_2. This makes the power

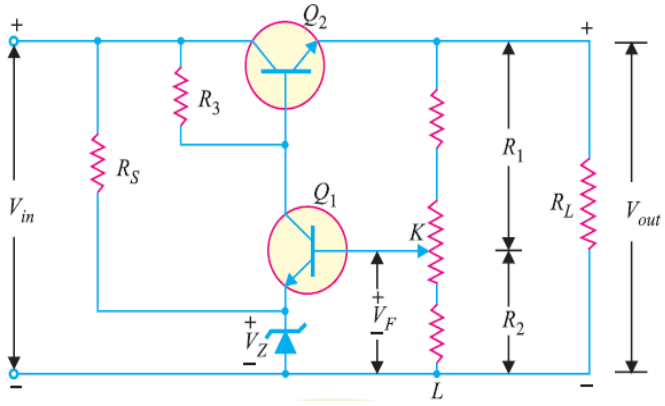


Fig. 2. A basic series pass transistor regulator circuit with voltage feedback.

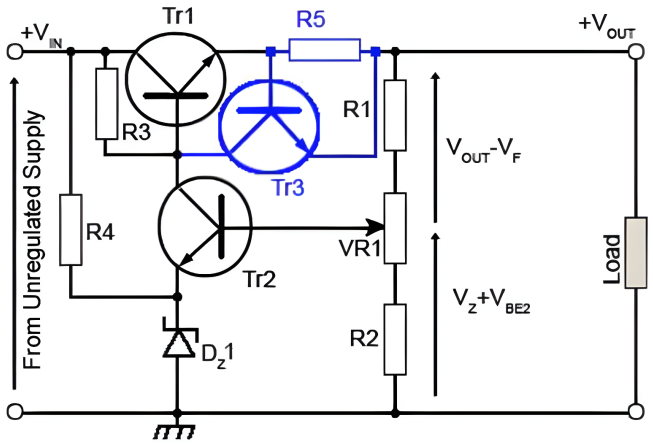


Fig. 3. A series pass transistor regulator circuit with voltage feedback and output current limiting capability.

transistor conduct less current to R_L and reduces the voltage across it, keeping the voltage across it regulated.

2) **The output voltage decreases**: Under this circumstance, the voltage across R_2 will decrease. This means that Q_1's base voltage is decreased, hence decreasing its collector current and the current across R_3. Furthermore, less current across R_3 means that the voltage drop across it will be lower, increasing the base voltage of Q_2. This makes the power transistor conduct more current to R_L and increases the voltage across it, restoring the regulation across it.

Another feature that can be added to the circuit for protection purposes is adding a current limiting transistor as shown in Figure 3. This will prevent damage to the supply in the event of too much current being drawn from the output, or even a complete short circuit across the output terminals. Two components have been added: $Tr3$ and R_5. The resistor R_5 is a very low value (typically less than 1 ohm). Its value can be determined by the following equation:

$$R_5 = \frac{V_{BE(Tr3)}}{I_{max}}$$

where $V_{BE(Tr3)}$ is the base-emitter voltage of the current limiting transistor, and I_{max} is the maximum current required across R_5. The maximum current required can be approximated as the output current (i.e the load current) since the current across R_1 and R_2 is usually very low compared to the output current.

When the load current exceeds a predetermined threshold, the voltage across R_5 increases to approximately 0.7V (assuming it is a silicon transistor), activating transistor $Tr3$. Because $Tr3$ is linked to the base-emitter junction of the primary control transistor $Tr1$, the activation of $Tr3$ decreases the base-emitter voltage of $Tr1$. The greater the excess current, the more $Tr1$ is restrained. This mechanism prevents the output current from exceeding the configured limit, even in the event of a short circuit. In such a case, the base-emitter voltage of $Tr1$ is reduced nearly to zero, which stops its conduction. Consequently, the output voltage falls to zero while the condition of excessive current persists, ensuring that the power supply remains secure and unharmed.

By incorporating all these features, the circuit ensures proper voltage regulation and safeguards the output load, maintaining reliable operation even under varying input supply conditions.

B. Components Selection

Our goal is to design a voltage regulator that outputs approximately 6.8-7.0 V with a voltage feedback and output current limiting capability. Therefore, component selection must be precise and carefully matched to ensure optimal performance, reliability, and safety of the circuit under all operating conditions.

Component selection was based on the following criteria, with this ensuring optimal performance and reliability of the circuit:

- **Components' availability**: building the circuit based on the available components in the market.
- **Voltage, current ratings, power dissipation and thermal limits**: components must handle the maximum expected operating conditions overheating should be prevented.
- **Efficiency**: minimizing energy loss.
- **Reliability and lifespan**: choosing durable and high-quality components. This can be done by ordering from reputable and trusted stores and shopping apps and websites.
- **Cost**: ensuring having cost-effective components while not comprising the quality of the design.
- **Compatibility with other components**: ensuring proper integration and stability in the system.

In the beginning, a 5.1 V Zener diode was chosen as a reference voltage since it is more available than other types of Zener diodes. We chose a type that has a power dissipation of 1W to ensure that it can handle high currents. The maximum current it can handle is:

$$I_{max} = \frac{P_Z}{V_Z} = \frac{1W}{5.1V} \approx 196mA$$

Moreover, the selected power transistor for the voltage regulator was **BD243C**; the reasons for picking it are as follows:

- 6A collector current.
- 100V V_{CE}.
- High power rating (65W).
- 2V base-emitter voltage V_{BE}.
- A common heatsink for thermal management (TO-220).

Furthermore, we chose **2SD880-Y** as the error amplifier transistor for the following reasons:

- 3A collector current.
- 60V V_{CE}.
- Adequate power rating (30W).
- 1V base-emitter voltage V_{BE}.

In addition, the current limiting transistor that was mainly selected based on the availability of transistors was **BD139** since it has:

- 1.5A collector current.
- 80V V_{CE}.
- Adequate power rating (12.5W).
- 1V base-emitter voltage V_{BE}.

The maximum current that we want in the output load is approximately 1.4 A since we want a 7V output for each dummy load resistor (5Ω, 10Ω, 15Ω, and 20Ω). Therefore, the maximum current will be produced by the minimum resistance value (5Ω), so the current is:

$$I_{max} = \frac{7V}{5\Omega} = 1.4A$$

For the current sensing resistor, the value of it will be the base-emitter voltage of the **BD139** transistor divided by the maximum output current:

$$R_{sensing} = \frac{V_{BE}}{I_{max}} = \frac{1V}{1.4A} \approx 0.714\Omega$$

and due to the availability of components in the market, the available resistor value is 0.5Ω. Moreover, it is important to have a resistor that can handle an amount of 1.4 A current. Therefore, it is important to put a safety margin of at least 2× the maximum current. The safe power rating for this resistor is:

$$P_{0.5\Omega} = (2I_{max})^2 \times R = (2 \times 1.4)^2 \times 0.5 = 3.92W$$

Based on the availability in the market, we bought a 5W 0.5Ω resistor.

For the voltage divider part of the circuit (R_1 and R_2), as previously shown in Figure 3, we chose the value for R_1 as 100Ω to have a simpler analysis for R_2, and since it is a commonly available resistor. For R_2, the voltage across it is determined by this formula:

$$V_{R_2} = V_F = V_Z + V_{BE(2SD880-Y)}$$

Substituting the values:

$$V_{R_2} = 5.1 + 1 = 6.1V$$

Now, to obtain the best value for R_2 that will maintain an output voltage of 7V , we will use the inverse voltage divider rule shown below and substitute the values we have.

$$V_{out} = V_F \times \frac{R_1 + R_2}{R_2}$$

$$7V = 6.1 \times \frac{100 + R_2}{R_2}$$

The value for R_2 that will maintain an output voltage of 7V is:

$$R_2 \approx 680\Omega$$

due to the variations in real-life experiments, it is preferable to have a potentiometer for tuning the output voltage.

Furthermore, a suitable power for the 100Ω can be known by knowing the voltage across it, which will be:

$$V_{100\Omega} = V_{out} - V_F$$

substituting the values:

$$V_{100\Omega} = 7 - 6.1 = 0.9$$

The power will be:

$$P_{100\Omega} = \frac{(V_{100\Omega})^2}{100\Omega} = \frac{0.9^2}{100} = 0.0081W$$

Therefore, a 1W 100Ω resistor will be more than enough to operate properly.

For R_3, the value of it can be determined by knowing the base voltage of the power transistor and assuming an adequate amount of current that goes to it to make the power and the error amplifier transistors operate properly. Since the base-emitter voltage V_{BE} of the power transistor is approximately 2V, the base voltage can be determined by knowing the emitter voltage, which will be approximately the output voltage (7V) due to the negligible value of the voltage drop in the current sensing resistor. Therefore:

$$V_{BE} = V_B - V_E$$

Substituting the values:

$$2V = V_B - 7$$

$$V_B = 9V$$

Let's assume that the current through R_3 is 50mA; the value of it based on that will be:

$$R_3 = \frac{V_{in} - V_B}{I}$$

Substituting the maximum value of the input voltage, V_B, and the value of the current:

$$R_3 = \frac{30 - 9}{50mA} = 420\Omega$$

based on the availability in the market, the best value will be 430Ω. To find a proper power rating for the resistor, we will put a 2× safety margin for the current, which means that the assumed value will be 100mA. The proper power rating will be:

$$P_{430\Omega} = I^2 \times R = 0.1^2 \times 430 = 4.3W$$

And based on the availability in the market, we bought a 430Ω 7W resistor.

For R_4, the value of it can be determined by assuming an adequate amount of current that goes through the Zener diode. The value of it will be:

$$R_4 = \frac{V_{in} - V_Z}{I}$$

For most Zener diodes, 20mA will be more than enough for it to operate properly. Substituting the values in the equation:

$$R_4 = \frac{30 - 5.1}{20mA} \approx 1.25k\Omega$$

For voltages stabilization, capacitors were used to achieve that goal in the circuit. In general, higher capacitance value means more stabilization. For the input voltage, a 100μF capacitor was connected in parallel with input voltage. We chose its voltage rating to be 60V for 2× safety margin. Therefore, we bought a 100μF capacitor with 63 V rating due to the availability of components.

Furthermore, more capacitors were placed with R_4 for more stabilization in the voltage and the current across the Zener diode. R_4 was divided into 3 resistors based on the availability of components:

- 2× 430Ω resistors (the same resistor for R_3).
- 220Ω resistor rated 1W (since the value of the current across it sequared ($I^2 = 0.02^2 = 0.0004$) will be very low, leading to less needed power rating).

The capacitors used were two 10μF capacitors that have a voltage rating of 30V. These capacitors provide local filtering at each step of the resistor divider, reducing noise and improving the stability of the voltage reference.

Table I shows the components used for building the voltage regulator.

TABLE I
COMPONENTS LIST

Component	Purpose / Function
V_{in} (DC Source)	Provides unregulated input voltage for the regulator
C_1 (100μF)	Input filtering to reduce ripple and noise
R_1 (430Ω)	RC filtering and pre-bias network for reference
R_2 (430Ω)	Additional filtering and stabilization for reference
C_2 (10μF)	Reference voltage filtering
C_3 (10μF)	Additional reference smoothing to prevent oscillations
R_3 (220Ω)	Zener diode bias resistor
5.1V Zener diode	Provides stable reference voltage
Q_1 (2SD880-Y)	Error amplifier transistor
Q_2 (BD243C)	Power transistor
Q_3 (BD139)	Current limiting transistor
R_5 (0.5Ω)	Current-sense resistor for over current protection
Potentiometer	Output feedback resistor for voltage regulation
R_7 (100Ω)	Additional feedback resistor stabilizing output
TO-220 Heatsink	Dissipates heat from the power transistor (Q_2)
R_L	Load resistor for testing regulated output

C. Simulation

To verify the calculations experimentally, we built the circuit on LTspice and ran a DC sweep from 10 V to 30 V as shown in Figure 4. The value of R_6 was altered to 470Ω in the simulation since the transistors in LTspice have different characteristics than transistors in real life. This change shows the importance of having a potentiometer for better regulation instead of a regular resistor.

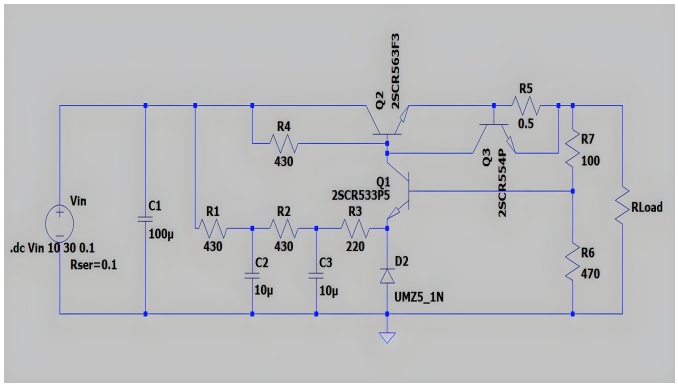


Fig. 4. The voltage regulator circuit built on LTspice.

Some tests were performed to to ensure that the circuit works properly. The tests are as follows:

1) **Open-Loop Test**: By disconnecting the load resistor R_L, we put a sweep input from 10 V to 30 V to record the voltage regulation across the output terminals. This method focuses on the fundamental actions of the regulator and enables us to monitor how the output voltage reacts only to changes in the input power supply. The measured voltage output shown in Figure 5 shows that the output voltage remains approximately constant even when a 20 V DC sweep from 10 V to 30 V is applied to it; it shows a minor change from 6.92 V to 7.04 V. This clearly shows the circuit's capability to uphold a consistent output when faced with varying

input situations. Let's take a 16V input as our DC operating point; the voltage output at this point is 6.96V. We will use this value to find the voltage regulation percentage at each load that will be connected.

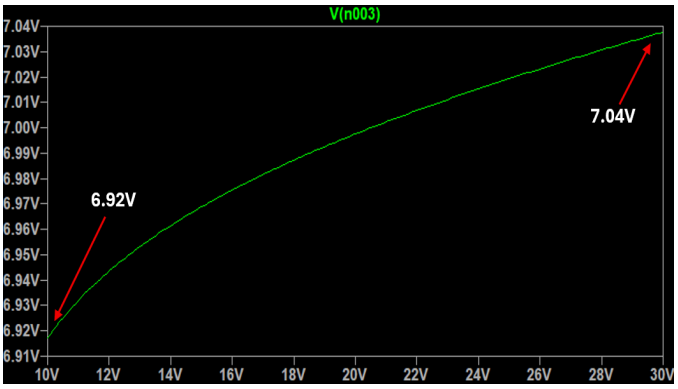


Fig. 5. The output voltage when no load is connected.

2) **Load Testing with Resistors**: For this part, we tested 4 load resistors R_L:
   - 5Ω resistor.
   - 10Ω resistor.
   - 15Ω resistor.
   - 20Ω resistor.

   These values were selected to represent a wide range of load conditions, from heavy loading (5 Ω, drawing the highest current) to relatively light loading (20 Ω, drawing the lowest current). By sweeping the input voltage and observing the corresponding output voltage and current, we were able to assess the regulator's ability to maintain a stable output near the desired 7 V level.
   This test is crucial because real-world electronic circuits rarely operate under a single fixed load; instead, the load often changes depending on the connected devices or operating state. A well-designed voltage regulator must therefore sustain a consistent output voltage despite these variations.
   The percentage voltage regulation of the circuit was evaluated using the following equation:

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

where V_{NL} represents the no-load output voltage and V_{FL} denotes the full-load output voltage. This formula allows us to quantify how much the output voltage changes when the load is applied.
Another factor for evaluating the performance of the voltage regulator is the efficiency, which is calculated using this law:

$$\eta = \frac{P_o}{P_{in}} \times 100$$

where:

$$P_o = V_o I_o \quad \& \quad P_{in} = V_{in} I_{in}$$

The output for each case was as follows:

a) 5Ω **resistor**: at this load, the output is observed to have a big sweep when varying the input voltage as shown in Figure 6. This is because the increased current that will be delivered to the load resistor R_L will cause the current limiting transistor to start operating, leading to variations in the output voltage.
Let's take 14 V input voltage as our DC operating point. The measured input current, output current, and output voltage are: 1.25 A, 1.23 A, and 6.15 V, respectively. Therefore, the efficiency for this circuit is:

$$\eta = \frac{6.15 \times 1.23}{14 \times 1.25} \times 100 \approx 43.2\%$$

The voltage regulation percentage is:

$$\%VR = \frac{6.96 - 6.15}{6.15} \times 100 \approx 13.2\%$$

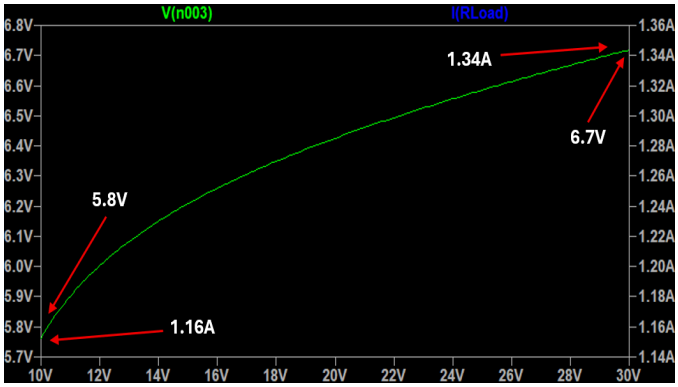


Fig. 6. The output voltage and current when the load is 5Ω.

b) 10Ω **resistor**: at this load, we can see that the output voltage and current are being more regulated as shown in Figure 7. This is because the current delivered to the output is less than the first case (5Ω resistor), which means that the output voltage will be more stabilized and the effect of the current limiting transistor is negligible in this case.
Let's take 14 V input voltage as our DC operating point. The measured input current, output current, and output voltage are: 727 mA, 695 mA, and 6.95 V, respectively. Therefore, the efficiency for this circuit is:

$$\eta = \frac{6.95 \times 0.695}{14 \times 0.727} \times 100 \approx 47.5\%$$

The voltage regulation percentage is:

$$\%VR = \frac{6.96 - 6.95}{6.95} \times 100 \approx 0.14\%$$

c) 15Ω **resistor**: when testing this load, we can see that the output voltage is still regulated at

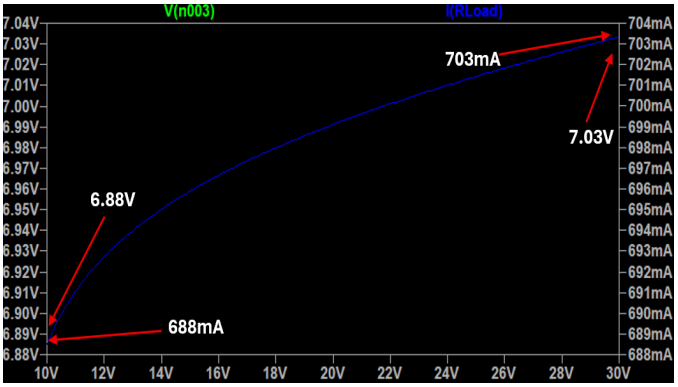


Fig. 7. The output voltage and current when the load is 10Ω.

approximately 7 V with minimal variations as shown in Figure 8. Generally, the higher the load resistance value, means less current is derived to the output, hence increasing the stability of the output voltage. While minor variations can be seen at the far ends of the load sweep, the regulator reliably upholds its standard output level. This behavior demonstrates that the circuit functions effectively within its intended regulatory limits. Additionally, the steady decrease in load current alleviates pressure on the pass elements, aiding in enhanced thermal efficiency and overall stability.
Let's take 14 V input voltage as our DC operating point. The measured input current, output current, and output voltage are: 493 mA, 466 mA, and 6.99 V, respectively. Therefore, the efficiency for this circuit is:

$$\eta = \frac{6.99 \times 0.466}{14 \times 0.493} \times 100 \approx 47.1\%$$

The absolute value of the voltage regulation percentage is:

$$\%VR = \frac{6.96 - 6.99}{6.99} \times 100 \approx 0.43\%$$

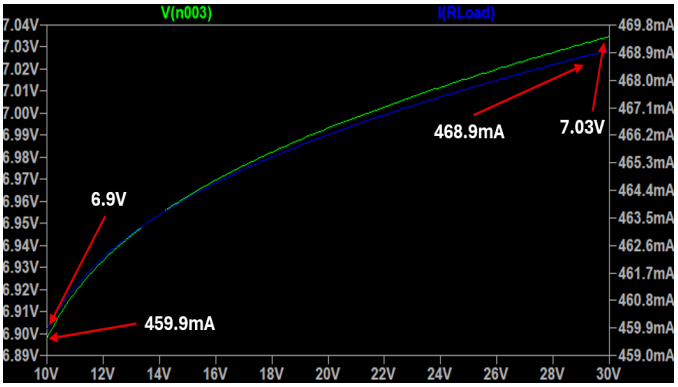


Fig. 8. The output voltage and current when the load is 15Ω.

d) 20Ω **resistor**: when this load is tested, the output voltage is more regulated and stabilized at 7 V due to reduced current delivered to the load by the circuit as shown in Figure 9. Let's take 14 V input voltage as our DC operating point. The measured input current, output current, and output voltage are: 380 mA, 350 mA, and 6.99 V, respectively. Therefore, the efficiency for this circuit is:

$$\eta = \frac{6.99 \times 0.35}{14 \times 0.38} \times 100 \approx 46\%$$

The absolute value of the voltage regulation percentage is:

$$\%VR = \frac{6.96 - 6.99}{6.99} \times 100 \approx 0.43\%$$

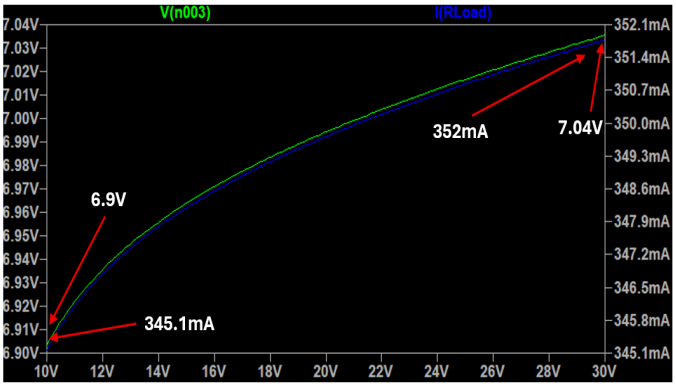


Fig. 9. The output voltage and current when the load is 20Ω.

The values of efficiency for each load are shown in Table II. Efficiencies in linear voltage regulators are usually low. Moreover, efficiency isn't the most important factor in a linear regulator; this is because the series pass transistor continuously dissipates the voltage difference between input and output as heat while maintaining approximately a constant output voltage.

TABLE II
MEASURED EFFICIENCY FOR DIFFERENT LOAD CONDITIONS

Load	Efficiency (%)
5 Ω	43.2
10 Ω	47.5
15 Ω	47.1
20 Ω	46.0

3) **Connecting a battery**: To test the circuit with a battery, we replaced the load with a voltage source in series with a small resistance, typically 2Ω as shown in Figure 10. To verify that the battery is charging, we will put a value under 6 V (5.7 V for example), and check if there is a current running in the battery.
As shown in Figure 11, there is a current that is going through the battery, indicating that the battery is charging.

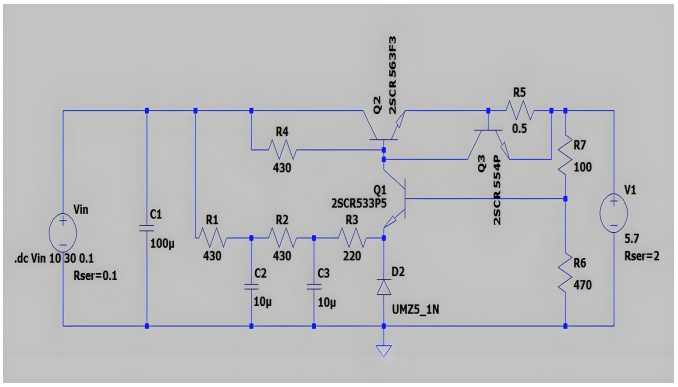


Fig. 10. Connecting a voltage source in series with a small resistance to simulate a battery.

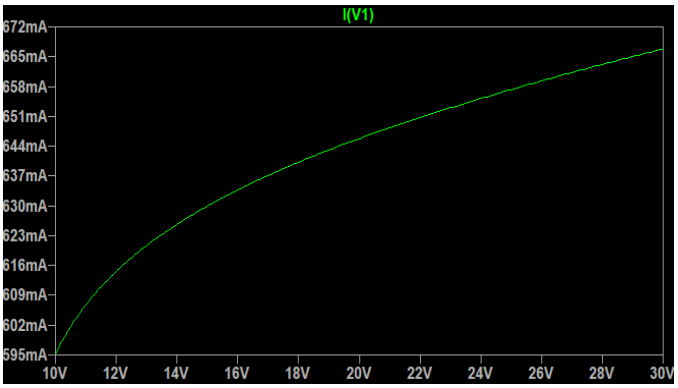


Fig. 11. The current through the battery (battery is charging).

TABLE III
MEASURED OUTPUT VOLTAGE AND CURRENT FOR DIFFERENT LOAD CONDITIONS

Load	Voltage (V)	Current (A)
No load	6.92–7.04	0
5 Ω	5.8–6.7	1.16–1.34
10 Ω	6.88–7.03	0.688–0.703
15 Ω	6.90–7.03	0.460–0.469
20 Ω	6.90–7.04	0.345–0.352

Table III shows a summary of the output results.

Using the values in this table, the output resistance can be computed using this law:

$$R_o = \frac{\Delta V_o}{\Delta I_o}$$

As shown in Table IV, the values of the output resistance are compared with the original values of the load resistances.

These tests show that the voltage regulator works properly under varying load and input voltage conditions, and is proper for charging a lead-acid battery.

D. Hardware Implementation

For hardware implementation, we design a printed circuit board (PCB) for the voltage regulator using EasyEDA, an online designing tool, to construct our circuit. The schematic diagram of the circuit is shown in Figure 12.

TABLE IV
OUTPUT RESISTANCE AND ERROR FOR DIFFERENT LOAD CONDITIONS

Load	Output Resistance R_o (Ω)	Error (%)
5 Ω	5	0
10 Ω	10	0
15 Ω	14.44	3.7
20 Ω	20.29	1.45

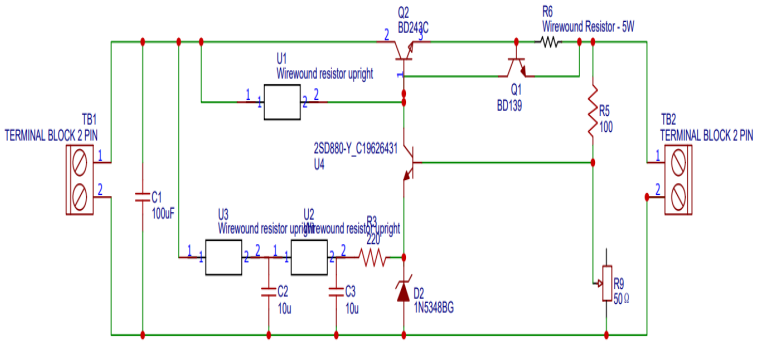


Fig. 12. The schematic diagram of the circuit.

To give the circuit an input voltage and observe the output from it, we used two 2.54mm terminal blocks in the input and output terminals to connect them with wires.

Figure 13 shows the design of the PCB and the corresponding connections. The input and output terminal blocks were placed at the edges of the PCB to ensure that all external wires remain easily accessible, reducing mechanical stress on the board and improving overall ease of installation during testing and operation.

Moreover, the components were arranged in a layout that closely follows the schematic diagram, minimizing the risk of confusion during assembly and ensuring that each component can be identified and placed correctly. In addition, thicker traces were used in the high-current output path leading to the load terminal (TB2) to guarantee safe current handling and enhance thermal performance.

To verify that the PCB layout was suitable for physical construction, a 3D model of the board was generated, as shown in Figure 14. This allowed us to visualize the final assembly and confirm that the component placement, spacing, and mechanical structure were appropriate. After validating the design, we moved on to the next step: building the circuit in real life.

Figure 15 shows the PCB Implementation and component overview.

III. RESULTS AND DISCUSSION

To verify that the circuit works properly under real-life conditions, several tests were performed to show the reliability

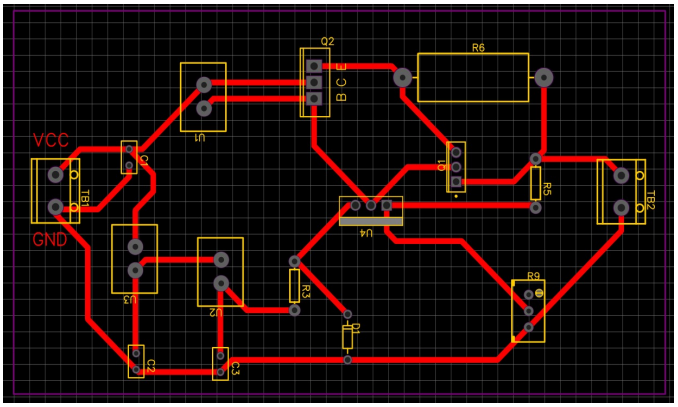


Fig. 13. The PCB design of the voltage regulator.

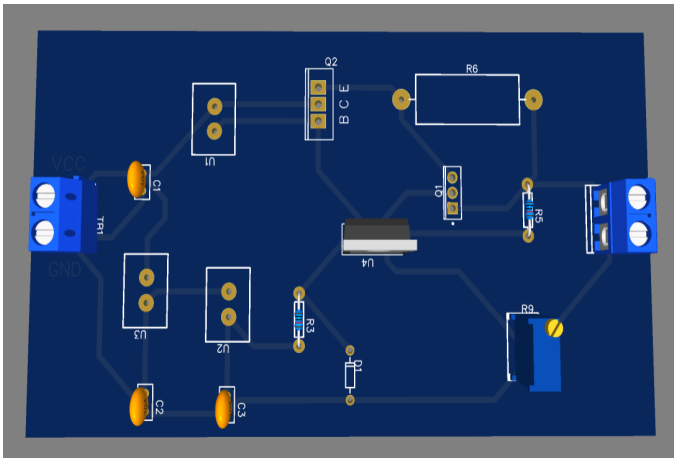


Fig. 14. The 3D view of the PCB design.

of the voltage regulator to charge the lead-acid battery. The tests include:

- **Open-Loop Test**: This initial test was performed to observe the natural behavior of the circuit without any load connected. By measuring the output voltage while sweeping the input supply, we ensured that the regulator produced a stable and predictable output around the desired 7 V level. The open-loop test is essential for confirming that the reference voltage, feedback network, and pass transistor are all functioning correctly before connecting any external load.
- **Load Testing with Resistors**: In this stage, different resistive loads were connected to simulate various output current conditions. This test demonstrates how effectively the regulator maintains its output voltage as the load changes. It also reveals the influence of current limiting and highlights the point at which the regulator begins to drop voltage due to increased current demand. Overall, this test verifies the regulator's dynamic response and stability under different operating conditions.
- **Connecting a Battery**: Finally, the regulator was tested under real charging conditions by connecting it directly

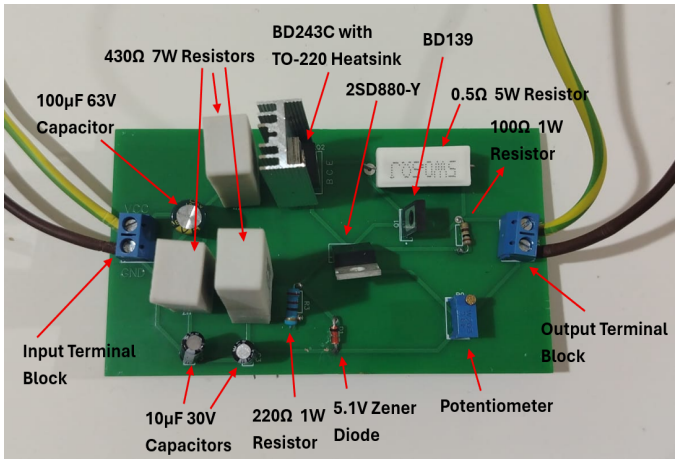


Fig. 15. The PCB Implementation and component overview.

to a lead-acid battery. This test is critical because a battery behaves differently from a simple resistive load: its voltage varies as it charges, and it draws current based on its state of charge. By monitoring both the input and output parameters during this process, we confirmed that the regulator can safely supply charging current while preventing overvoltage. This step validates the practical usefulness of the circuit and ensures that it performs reliably in its intended application.

A. *Open-Loop Test*

We tested this case with varying input conditions, including:

- 10 V input.
- 20 V input.
- 25 V input.
- 30 V input.

These points were used to verify that the voltage regulator performs well under varying input conditions. Figure 16 shows the output at the different input voltages; it can be seen that there is a minimal variation in the output voltage from 6.78 V to 7.01 V, which means that the voltage is regulated. Comparing these values to simulation results, there is a slight difference in the output voltage due to tolerance in components. Overall, the results show that we no load is connected (R_L=$\infty\Omega$), the output voltage will be more regulated since less current is delivered means more stabilization. At an input voltage of 16 V, the output voltage is 6.85 V. We will use this value to determine the voltage regulation percentage of the loads.

B. *Load Testing with Resistors*

Due to the availability of components in the market, we were only able to find a 20Ω resistor with 10 W power rating. Other loads were not found since they are very limited in the market. We tested this case with varying input conditions, including:

- 10 V input.
- 20 V input.

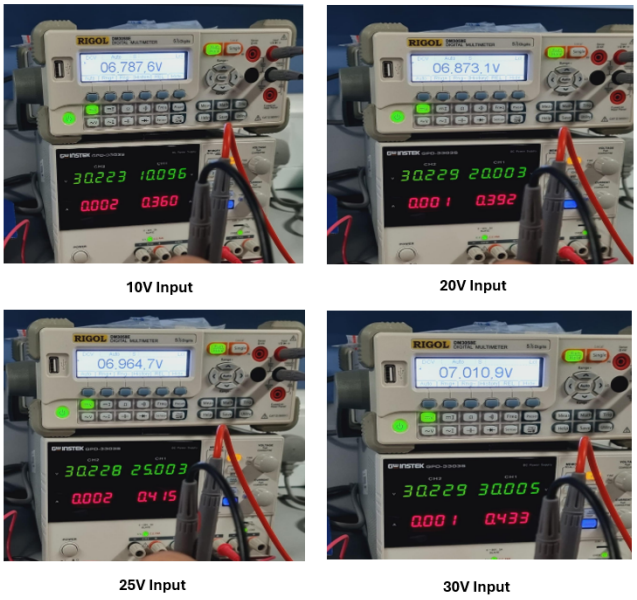


Fig. 16. Various tests' results when no load is connected.

- 30 V input.

These points were used to verify that the voltage regulator performs well under varying input conditions.

Figure 17 shows the output voltage for each condition, which shows that the output voltage has small variations (6.79-6.99 V) and high performance in regulation. Comparing these results to simulation results, we can see that the real-life experiment has more variations. The reason behind that is due to tolerance in the components.

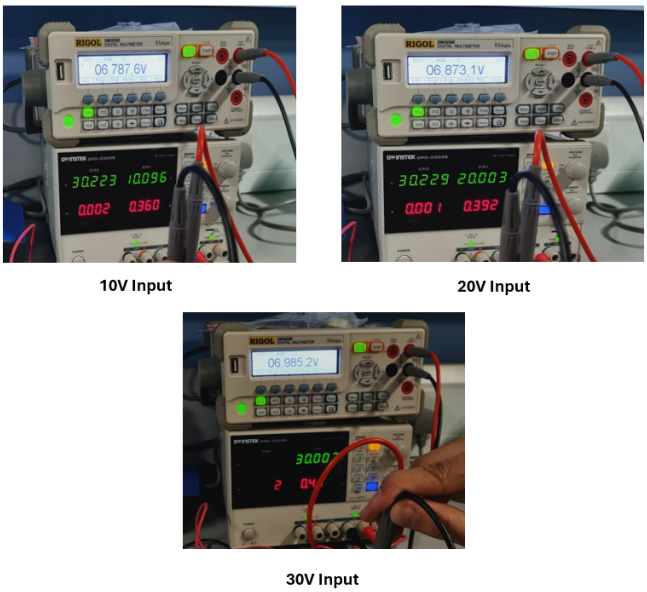


Fig. 17. Various tests' results when the 20Ω load is connected.

Let's take 16 V input voltage as our DC operating point. The measured input current, output current, and output voltage are: 379 mA, 340 mA, and 6.82 V, respectively, as shown in Figure 18. Therefore, the efficiency for this circuit is:

$$\eta = \frac{6.82 \times 0.34}{16 \times 0.379} \times 100 \approx 40\%$$

Linear voltage regulators are usually inefficient because they control voltage by converting excess energy into heat. Any difference between input and output voltage is wasted as power loss, making them much less efficient than switching regulators, especially when the voltage drop is large.

Moreover, the output resistance is calculated to be 20.06Ω, with an error percentage of 0.3%.

In addition, the voltage regulation percentage is approximated to:

$$\%VR = \frac{6.87 - 6.82}{6.82} \times 100 \approx 0.73\%$$

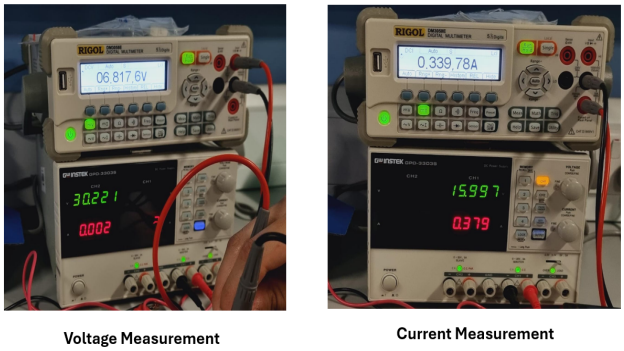


Fig. 18. Voltage and current measurements at an input voltage of 16 V.

C. Connecting a Battery

For this part, we will connect a discharged 6 V lead-acid battery to the output terminal and check if there is a current running to the battery to ensure that it is charging. This test is essential since a battery behaves differently from a simple resistive load: its voltage varies as it charges, and it draws current based on its state of charge.

In the beginning, we check if the battery is discharged by measuring its voltage. As seen in Figure 19, we can see that the voltage is under 6 V, which verifies that it is discharged. Moreover, we connected the battery with the output wires and measured the current that is going to the battery to verify if it's charging. This was done by connecting the leads of the multimeter in series with the battery. As shown in Figure 20, there is a current that goes through the battery, indicating that our design has succeeded in charging a 6 V lead acid battery. The values observed in the experiment are different than what is observed in the simulation since the battery has different characteristics than just a simple voltage source with a small impedance in series.

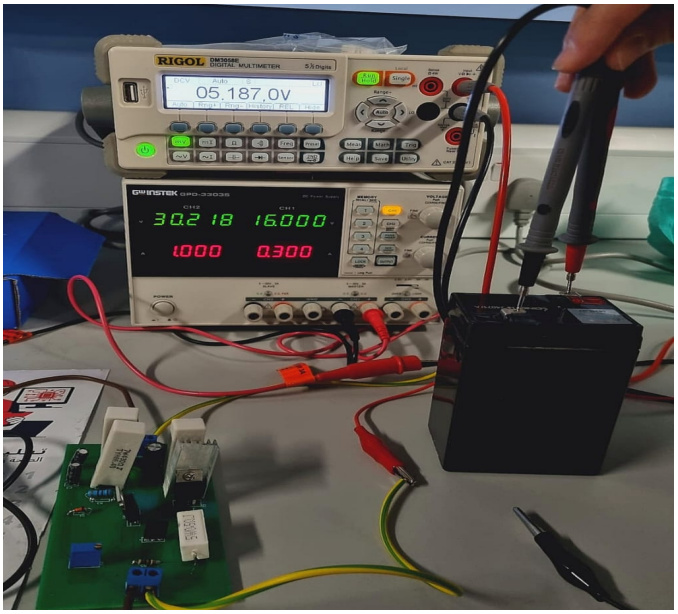


Fig. 19. Checking the battery status (charged/uncharged).

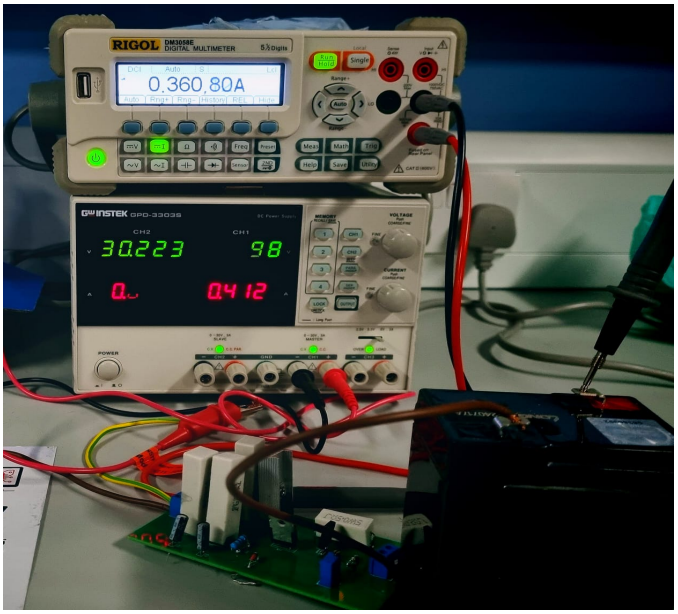


Fig. 20. Charging the battery using the voltage regulator

D. Discussion

The results of the experiments show that the voltage regulator works reliably and performs as expected under real-life conditions. In the open-loop test, the output voltage remained close to 7 V even when the input voltage was changed from 10 V to 30 V, which confirms that the reference network and pass transistor regulate the voltage properly. When testing with a 20 Ω load, the regulator continued to maintain a stable output, although the measurements showed slightly more variation compared to the simulation. This is mainly due to real-world factors such as component tolerances and internal losses. At 16 V input, the efficiency was around 40%, which is normal for a linear regulator where the extra voltage is dissipated as heat. Finally, connecting a discharged lead-acid battery confirmed that the regulator can safely charge it, since a measurable charging current was observed. This demonstrates that the circuit not only regulates voltage well but is also suitable for practical battery-charging applications. Overall, the tests verify that the design is stable, functional, and capable of delivering the required performance.

IV. Conclusions

A. Summary

This project successfully demonstrated the design and hardware implementation of a 6 V lead-acid battery charger based on a discrete series pass regulator. The regulator was designed mainly using a Zener reference, a power transistor, an error amplifier transistor, and a current limiting transistor. The regulator functioned and maintained a stable and reasonable voltage within the expected range of 6.8-7.1V. We ensured proper component selection based on the availability, quality of components, power ratings and thermal management, and cost to ensure better design and regulation. Our LTSpice simulation verified the expected output and the functioning of our design before the hardware construction, confirming the safe operating point across the full input voltage range of 10–30 V.

We tested our complete charger circuit on real hardware and real loads, and the results showed that the design works correctly and matches the main simulation results. When we connected the 6V lead-acid battery, we observed that the battery was charging by ensuring that there was current going to the 6V lead-acid battery. We observed good voltage regulation, output resistance, and efficiency of the regulator from the measurements. The observed efficiency followed the expected behavior of a linear regulator at a certain DC operating point. The circuit did not show any sign of unstable charging, overvoltage, or dangerous overheating, which might have led to burning the circuit.

In addition to the technical and engineering objectives, this project helped us a lot in understanding how such circuits function and operate, and also by understanding the feedback control, thermal considerations in the hardware implementation, and ensuring proper connections of the transistors based on the datasheets. In the end, our final design met all the required specifications and provided a good foundation in understanding the world of electronics, batteries, and their applications.

B. Future Improvements

Things that can be done in the future for better and enhanced performance of the circuit include:

- **LED Indicators**: We can add LEDs to indicate the state of charging the battery. For instance, we can connect a red LED to indicate that the battery is still charging, and a green LED to indicate that the battery is fully charged.

- **Adding Thermal Shutdown.**: An important improvement for the voltage regulator design is the implementation of a thermal shutdown feature. This protection mechanism automatically turns off the regulator when the junction temperature exceeds a safe threshold, preventing damage to the circuit components due to overheating. Incorporating thermal shutdown enhances the reliability and longevity of the device, especially under high-load or fault conditions, ensuring safe operation without requiring constant manual monitoring. This feature is particularly valuable in applications where the regulator may be exposed to varying ambient temperatures or prolonged periods of high current draw.
- **Over-Voltage Protection**: Some additional features, such as over-voltage protection, can be used to enhance the stability of the output voltage in our circuit. Figure 21 shows how an over-voltage protection feature can be added to the voltage regulator as an additional feature.

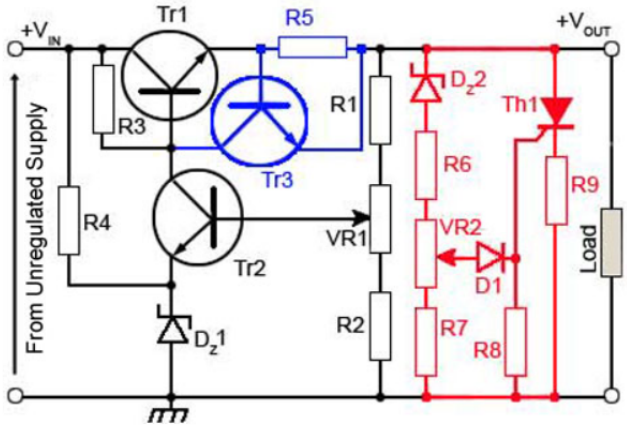


Fig. 21. A voltage regulator with overcurrent and overvoltage protection.

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